Howard (Hao) Ye

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LinkedIn — GitHub — Portfolio

SUMMARY

Results-driven DevOps/DevSecOps Engineer with hands-on experience building automated CI/CD pipelines, secure containerized microservices, resilient cloud infrastructures. Adept at infrastructure automation (Ansible, Terraform), application security (RBAC, JWT, rate limiting), and deploying full-stack, AI-integrated platforms. Proven ability to bridge software engineering and IT operations to deliver scalable, production-ready solutions.

TECHNICAL SKILLS

CI/CD & Automation: GitHub Actions, Jenkins, Ansible, Terraform, Molecule
Containers & Cloud: Docker, Kubernetes, AWS (EC2, RDS, S3), GCP, Nginx, Render, Vercel
DevSecOps & Security: RBAC, JWT, UFW, fail2ban, WireGuard VPN, Rate Limiting, API Key Auth
Programming: TypeScript, JavaScript, Python, Node.js, C/C++, React, Vue.js, Bash
Databases: PostgreSQL, Redis, SQLite
Observability: Prometheus, Grafana, Alertmanager, Node Exporter
Testing: Jest, Supertest, Molecule, Shell scripting (API testing)

EDUCATION

MS Computer Engineering — University of Tennessee Knoxville — 2025

EXPERIENCE

Volunteer AI Software Developer — MatchingDonors.com (Remote) Jan 2026 – Present
Full-Stack Healthcare Platform with CI/CD, Admin Portal & Mobile Support — [Repo] — [Demo]

- Building a multi-role organ donor matching platform (React/TypeScript, Node.js/Express) serving donors, patients, sponsors, and advertisers; features AI-powered article search, real-time Socket.IO chat, automated Resend API email notifications, and a React Native mobile client.
- Implemented enterprise-grade CI/CD pipeline: GitHub Actions for automated backend testing on every push; multi-stage Jenkins pipeline covering checkout, dependency installation, and QA test execution.
- Designed and enforced Role-Based Access Control (RBAC) with JWT authentication across 9+ RESTful API route modules (admin, auth, chat, content, profiles, notifications); wrote integration tests covering security and admin scenarios with Jest and a shell-based API test suite.
- Integrated Google Gemini API for AI-driven content crawling, article categorization, and a conversational chatbot assistant; utilized SQLite for portable, zero-config data persistence.

Volunteer AI Software Developer — MatchingDonors.com (Remote) Mar 2026 – Present
Social Media Automation Agent — [Repo]

- Engineered an AI-powered social media agent that autonomously crawls live transplant news, generates multi-post Bluesky threads via Google Gemini, and publishes through the AT Protocol API.
- Built an interactive CLI review loop with draft editing, regeneration, and auto-condensing via Gemini; integrated CronService for scheduled posting and HistoryService to deduplicate published content.
- Implemented URL shortening, character-count validation, and a modular TypeScript service architecture (crawler, AI, social, URL services) to support extensible content pipelines.

PROJECTS

Linux Server Hardening & Observability Automation — [Repo]
Tech: Ansible, Terraform, GCP, Prometheus, Grafana, Alertmanager, Node Exporter, Docker Compose, WireGuard, Certbot, UFW, fail2ban, GitHub Actions, Molecule

- Engineered a modular 8-role Ansible project (base, nginx, certbot, ssh, monitoring, blue-green, wireguard, app) to fully automate provisioning of a GCP Ubuntu server from bootstrap to production.
- Hardened the server with SSH key-only auth, UFW default-deny firewall, and fail2ban brute-force protection; restricted exposed ports to 22/80/443/51820 only.
- Deployed a containerized observability stack (Prometheus + Grafana + Alertmanager + Node Exporter) via Docker Compose; wrote 3 custom Prometheus alert rules for CPU > 80%, memory > 85%, and service

downtime with 1-minute resolution windows.

- Provisioned WireGuard VPN (multi-peer) and automated Let's Encrypt SSL/TLS issuance and 90-day renewal via Certbot; implemented blue-green deployment with automated traffic switching for zero downtime.
- Enforced playbook quality via GitHub Actions (ansible-lint on every push) and validated role idempotency with Molecule integration tests; infrastructure-as-code also supported by Terraform configurations.

Clinical Data Reconciliation Engine — AI-Powered Healthcare API — [Repo] [Demo]

Tech: Node.js/TypeScript, React, Docker, SQLite, Google Gemini 2.5 Flash, Jest, Render

- Designed and deployed an AI-powered medication reconciliation REST API resolving conflicting records from multiple healthcare systems (EHR, pharmacy, patient portal) with confidence scoring (0–100%).
- Containerized the full-stack application with Dockerfile and Docker Compose; deployed to Render.com with health check endpoint, environment variable management, and persistent SQLite via Docker volume.
- Added production hardening: LLM response caching (˜60% cost reduction), rate limiting (30 req/15 min), API key authentication, and full audit trail logging; wrote Jest/Supertest unit and integration tests.

Cloud-Native HPC Simulation Platform — [Repo]

Tech: Docker, Node.js/TypeScript, PostgreSQL, Redis, AWS EC2

- Architected a 4-service microservices platform with Docker Compose orchestration and health checks; deployed to AWS EC2 with security groups and monitoring.
- Built RESTful API gateway with JWT authentication, Redis caching, and PostgreSQL query optimization; implemented comprehensive test suite covering 15+ endpoints.

Smart Home Energy Management System — [Repo] — [Demo]

Tech: Docker, Vue.js/TypeScript, Node.js, Python, PPO (Reinforcement Learning), Socket.IO

- Built event-driven microservices with 4 containerized services processing real-time data from 11+ IoT devices; integrated a reinforcement learning service for automated device scheduling optimization.
- Implemented multi-environment deployment: local Docker development and Vercel production.

HPC Cluster Performance Simulator (MS Research) — [Repo]

Tech: C/C++, OMNeT++, Distributed Systems, Lustre

- Developed a discrete-event simulation framework in C++/OMNeT++ to model large-scale HPC storage networks with Fattree topologies and Lustre parallel file systems.
- Engineered automated benchmarking tools and data analysis pipelines for distributed computing performance evaluation.

Applied AI/ML Techniques Across Kaggle Challenges

Tech: PyTorch, Hugging Face Transformers, XGBoost, LightGBM, CatBoost, Optuna, scikit-learn

- **LLM Response Classification:** Fine-tuned DeBERTa-v3-base transformer for multi-class classification; custom TRL training pipeline with weighted F1-score optimization via hyperparameter tuning.
- **Risk Prediction System:** Ensemble gradient boosting with polynomial feature engineering; Optuna hyperparameter search over 100+ trials.
- **Bioinformatics Classification:** Multi-label protein function prediction using k-mer extraction and Gene Ontology hierarchy propagation.

ACADEMIC EXPERIENCE

MS Computer Engineering — University of Tennessee — 2025

- **Distributed Systems & HPC:** Researched file system performance optimization using machine learning, designing performance modeling frameworks and applying DevOps practices (Docker orchestration, CI/CD).